

RAHUL KUMAR JHA

Aspiring Machine Learning Engineer | AI & Deep Learning

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🔗 rahul5857 📁 Portfolio

Professional Summary

AIML undergraduate with hands-on experience in Machine Learning, Deep Learning, Natural Language Processing, and Retrieval-Augmented Generation (RAG) systems. Skilled in building AI-powered applications using FastAPI, Sentence Transformers, Docker, Kubernetes, and SQL.

Education

Bachelor of Technology (B.Tech) in Computer Science (AI-ML), <i>Malla Reddy Institute of Technology and Science</i> Expected Graduation: May 2027 CGPA: 8.9 / 10	08/2023 – 05/2027 Hyderabad, Telangana
Certification Program in AI-ML with Specialization in Drone Technology, IIT Hyderabad	06/2025 – 01/2026 Hyderabad, Telangana

Technical Skills

Programming Python (OOP, File Handling, APIs, Data Structures)	Machine Learning Regression & Classification, Model Evaluation, Cross Validation, GridSearchCV, Scikit-learn
Deep Learning Neural Networks, Backpropagation, ReLU, Sigmoid, Dropout	Natural Language Processing Sentence Transformers, Text Embeddings, Cosine Similarity, Semantic Search, RAG
Backend Development FastAPI, REST APIs	Data & Tools NumPy, Pandas, Matplotlib, Seaborn, Git, GitHub, Docker, Kubernetes, ChromaDB, LangChain, VS Code, Jupyter Notebook
Database SQL (CRUD Operations, Joins, Aggregations)	

Projects

AI Resume Analyzer FastAPI NLP Sentence Transformers Docker  - Built an AI-powered Resume Analyzer that compares resumes against job descriptions using NLP embeddings and cosine similarity. - Developed REST APIs using FastAPI for resume upload and automated analysis. - Implemented skill matching and missing skill identification based on job requirements. - Containerized the application using Docker and configured Kubernetes deployment files.
RAG PDF Chatbot LangChain ChromaDB Sentence Transformers FastAPI  - Built a Multi-PDF RAG chatbot using Python, Streamlit, ChromaDB, and Ollama for semantic document search and question answering. - Implemented PDF parsing, text chunking, embedding generation with Sentence Transformers, and vector similarity search. - Integrated Llama 3.2 for context-aware responses and metadata-based document filtering.
AI-Based Landing Zone Safety Classification AI-Assistant Python scikit-learn Pandas  - Designed a classification model to assess safe drone landing zones using image-derived features. - Performed data preprocessing, feature engineering, and model validation to ensure accurate and reliable predictions.